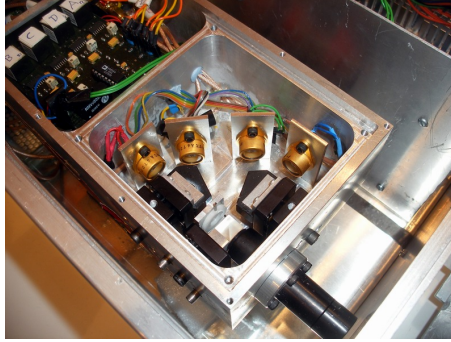


PHYSICS OF COMPLEX SYSTEMS

LECTURE AND TUTORIALS – PROF. DR. HAYE HINRICHSSEN – MAXIMILIAN SCHÄFFERER – SS 2026



Optical quantum cryptography device [Wikimedia]

EXERCISE 11.1: ENTANGLEMENT OF A PURE 2-QUBIT STATE (4P)

Consider the 2-qubit state $\rho = |\psi\rangle\langle\psi|$ with

$$|\psi\rangle = \frac{1}{5}|00\rangle + \frac{2i\sqrt{2}}{5}|01\rangle - \frac{4}{15}|10\rangle - \frac{8i\sqrt{2}}{15}|11\rangle.$$

- (a) Use *Mathematica*® or similar tools to compute the 4×4 matrix ρ in the qubit basis. Verify that ρ is pure. (2P)
- (b) Compute the reduced density matrices $\rho^{(L)}$ and $\rho^{(R)}$ for the left and the right qubit by carrying out the partial traces (1P)

$$\rho_{jk}^{(L)} = \sum_{m=0}^1 \rho_{jm,km}, \quad \rho_{mn}^{(R)} = \sum_{j=0}^1 \rho_{jm,jn}.$$

- (c) A pure state is called *entangled* if the reduced density matrices are mixed. Find out whether ρ is entangled or not. (1P)

EXERCISE 11.2: QUANTUM CRYPTOGRAPHY (8P)

Encrypted communication requires that both parties share a key (a random sequence of classical bits) that can be used to encode and decode messages by bitwise XOR operations. The problem of encryption therefore reduces to safely generate identical keys on both sides. In 1992 H.C. Bennett proposed a particularly simple protocol (B92):

Alice randomly generates a classical bit $a \in \{0, 1\}$, using it to prepare a qubit

$$|\psi\rangle = \begin{cases} |0\rangle & \text{if } a = 0 \\ |+\rangle & \text{if } a = 1 \end{cases}$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Then she sends this qubit to Bob.

Likewise Bob randomly generates a classical bit $b \in \{0, 1\}$. He uses this bit to select one of the two measurement apparatuses

$$\mathbf{M}_0 = |0\rangle\langle 0| - |1\rangle\langle 1|, \quad \mathbf{M}_1 = |+\rangle\langle +| - |-\rangle\langle -|$$

which is then used to measure Alice's incoming qubit. Then he sends the measurement result $m = \pm 1$ by classical public communication to Alice. Both Alice and Bob repeat the process until they get the result $m = -1$.

- (a) Show that $a = 1 - b$ if $m = -1$, that is, the locally generated bits are opposite. (2P)
- (b) How many qubits are on average needed to generate a bit of the key? (1P)
- (c) Suppose that Eve intercepts the communication in that she measures the transferred qubit by a projective measurement $\mathbf{E} = |\phi_+\rangle\langle\phi_+| - |\phi_-\rangle\langle\phi_-|$, where

$$|\phi_+\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\phi_-\rangle = \beta^*|0\rangle - \alpha^*|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

are two arbitrary orthogonal states. Show that Eve's interception generally changes the measurement statistics seen from the perspective of Bob and Alice, depending effectively on two parameters $u := |\alpha|^2$ and $v := \frac{1}{2}|\alpha + \beta|^2$. (3P)

- (d) Determine u and v such that Bob and Alice cannot detect the interception. (1P)
- (e) In the case of (d), is it possible for Eve to get information about the key? (1P)

($\Sigma = 12\text{P}$)

Almost done! Please submit your solution electronically as a single PDF file until Wednesday, July 01, 2026, via WueCampus.